Arithmetic Quantum Cat Maps:
Exact Fourth-Moment Reductions, Cyclotomic Row Support, and Heisenberg-Scale Trace Criteria

Maksim Churilov

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Abstract

We develop exact arithmetic reductions for a fourth-moment problem arising in quantum cat maps and isolate two logically separate closure problems before a microscopic Bohigas–Giannoni–Schmit (BGS) statement can be attempted. Let p be an odd prime, let $H \le \mathbf{F}_p^\times$ have odd prime order K, and put

$$c(h) = \frac{h+1}{h-1}, \qquad A = c(H \setminus \{1\}).$$

An exact pair-key reduction gives

$$B(t) = O(t) + 2(p-2)C_H(t) + (p-2)^2\mathbf{1}_{t=0},$$

and both nontrivial fourth-moment fluctuation layers are controlled by the additive energy $E_+(A)$. A source-exact subgroup-curve theorem yields

$$E_+(A) < 34K^{8/3} \qquad (100 < K < p^{3/4}/3),$$

while the corrected off/off elliptic analysis gives

$$\mathrm{Var}_{t\neq 0}\, O(t) \ll pE_+(A) + K^2p + K^4p^{1/2} + K^4.$$

The pointwise arithmetic obstruction admits a sharper exact geometry. A fixed-product decomposition reduces non-forced excess to intersections of special affine copies of the prime cyclotomic trace set. Fixed-degree subresultants in the maximal real cyclotomic field define an exact split-prime order-three support ${\Lambda'_K}^{(3)}$; outside it,

$$E_+(A) \le 11(K-1)(K-2).$$

Bass's prime-conductor relations for cyclotomic units imply a complete characteristic-zero affine-intercept classification: every intercept fibre has cardinality at most two, and all double fibres are explicit. Thus order-three intersections are genuinely created by finite reduction.

For a general row ratio we replace the many-coefficient subresultant support by one scalar shift resultant

$$\mathcal{R}_{K,r}(Z) = \mathrm{Res}_T(P_1(T), R_r(\alpha_r T + \beta_r + Z)) = c_0 + c_1 Z + c_2 Z^2 + \cdots,$$

whose order at $Z = 0$ is exactly the row-intersection number. Hence the event $I \ge 3$ is equivalent to simultaneous vanishing of c_0, c_1, c_2 modulo one split prime ideal. Individual intersections are also congruences between explicit products of cyclotomic units. For each fixed ratio, the supported split primes satisfy

$$\sum_{p\equiv 1\,(K),\ I_r(p)\ge 3} \log p = O(K^3).$$

The inverse ratio $r = -1$ has additional structure. If $F(X) = P(\tau + X)$ and x_i are its roots, then universally

$$\mathrm{Res}_X(F(X), F(Z - X)) = 2^m F(Z/2)\, S(Z)^2, \qquad S(Z) = \prod_{i<j}(Z - x_i - x_j),$$

with $m = (K-1)/2$. At split primes the diagonal factor is a unit at $Z = 0$, so $I_{-1} = 2\,\mathrm{ord}_0\, S$ and the order-three event is exactly $S(0) = S'(0) = 0$. Writing $F_Z(X) = F(X + Z/2) = E_Z(X^2) + XO_Z(X^2)$, we further prove

$$S(Z) = \pm\,\mathrm{Res}_Y(E_Z, O_Z).$$

Thus the inverse support is controlled by a half-degree resultant and its translation derivative, computable without the degree-m^2 shift resultant.

A companion minimal-extension theorem uses Frobenius: if $f = \mathrm{ord}_K(p)$ and the order-K subgroup is placed in $\mathbf{F}_{p^f}$, every non-forced collision orbit has size exactly f. Together with the global resultant budget this gives a density-one exact baseline for unrestricted prime pairs $K \asymp X^\theta$, $p \asymp X$ for every fixed $\theta < 1/3$; it does not improve the split slice, where $f = 1$.

Finally, partition-Möbius inversion and Poisson summation give an exact trace-to-local-statistics bridge. Microscopic band-limited k-point statistics require joint trace information at powers of order the matrix dimension, the Heisenberg scale. The missing spectral theorem is therefore a weighted Heisenberg-scale trace comparison for the correct desymmetrized block, plus a high-frequency tail estimate beyond band-limited tests. No microscopic BGS/GOE/COE universality theorem is claimed. Separately, a centered arithmetic normal form controls the actual cosine-kicked Pell propagator at its recurrence time. It gives a wrapped Gaussian law for its Kth power at quantum kick strength $\tau/\sqrt{2K}$, with an explicit normalized Hilbert–Schmidt comparison error. This concerns a vanishing classical perturbation and logarithmic recurrence times, not microscopic BGS.

1 Scope and evidence discipline

Throughout the split finite-field part, p is an odd prime and $H \le \mathbf{F}_p^\times$ has odd prime order K. Division is in $\mathbf{F}_p$. Universal claims are called exact only when the load-bearing derivation is present; finite computations are certificates, regressions or falsifiers, never substitutes for asymptotic proof. Literature inputs are used only with their hypotheses checked in the required regime.

The distinction between moment information and microscopic spectral statistics is essential. Arithmetic cat-map literature contains strong matrix-element and quantum-ergodicity results [1, 2, 3, 4, 5], while random-matrix universality is rigorous for special model classes such as mixing quantum graphs under their own hypotheses [6, 7]. None of these results is silently transferred to the arithmetic family considered here.

2 Exact pair-key geometry

For $x, y \in \mathbf{F}_p \setminus \{0, 1\}$ define

$$u = \frac{x}{y}, \qquad v = \frac{x-1}{y-1}, \qquad d = x - y. \tag{1}$$

Proposition 2.1 (Pair-key bijection)**.** *If $x \neq y$, then $u, v \in \mathbf{F}_p^\times \setminus \{1\}$, $u \neq v$, and*

$$y = \frac{1-v}{u-v}, \qquad x = \frac{u(1-v)}{u-v}, \qquad d = \frac{(u-1)(v-1)}{v-u}. \tag{2}$$

Conversely every such (u, v) gives a unique admissible off-diagonal pair. The diagonal collapses to $(u, v, d) = (1, 1, 0)$ with multiplicity $p - 2$.

Proof. The equations $x = uy$ and $x - 1 = v(y - 1)$ give $(u - v)y = 1 - v$, hence (2). The diagonal statement is immediate. □

Under the subgroup constraints $u, v \in H$, there are exactly $(K - 1)(K - 2)$ admissible off-diagonal ordered pairs. If $O(t)$ denotes the off/off contribution and

$$C_H(t) = \#\left\{(u, v) \in (H \setminus \{1\})^2 : u \neq v,\ \frac{(u-1)(v-1)}{v-u} = t\right\},$$

then exact bookkeeping gives

$$B(t) = O(t) + 2(p - 2)C_H(t) + (p - 2)^2\mathbf{1}_{t=0}. \tag{3}$$

The factor two is essential: diagonal/off and off/diagonal are distinct ordered classes. The total count is

$$|V_4| = (K^2 + 1)p^2 - (6K^2 + 3K)p + 13K^2 - 3K. \tag{4}$$

If $\Sigma_4(q) = \sum_t B(t)e_p(qt)$, $V = \sum_t B(t)$ and $Z = B(0)$, finite Fourier orthogonality gives

$$\mathbb{E}_{q\neq 0}\Sigma_4(q) = \frac{pZ - V}{p - 1}, \qquad \mathrm{Var}_{q\neq 0}\,\Sigma_4(q) = p\,\mathrm{Var}_{t\neq 0}\,B(t). \tag{5}$$

No probabilistic independence assumption enters this reduction.

3 Cayley energy and the best uniform bound

Put

$$c(h) = \frac{h + 1}{h - 1}, \qquad A = c(H \setminus \{1\}).$$

For $u \neq v$,

$$\frac{(u-1)(v-1)}{v-u} = \frac{2}{c(u) - c(v)}, \qquad c(h^{-1}) = -c(h). \tag{6}$$

Thus $A = -A$ and the nonzero multiplicities of C_H are additive-difference multiplicities of A up to the permutation $x \mapsto 2/x$. Hence

$$\sum_t C_H(t)^2 = E_+(A) - |A|^2, \qquad E_+(A) = \sum_x r_{A-A}(x)^2. \tag{7}$$

Identity and central inversion force the nonzero baseline

$$(K - 1)(2K - 5). \tag{8}$$

For the unnormalized Fourier transform,

$$E_+(A) = \frac{(K-1)^4}{p} + \frac{1}{p}\sum_{q\neq 0} |\widehat{1_A}(q)|^4. \tag{9}$$

The sharp pointwise target is therefore

$$E_+(A) = \frac{(K-1)^4}{p} + O(K^{2+o(1)}). \tag{10}$$

For $r_A(t) = \#\{(a, b) \in A^2 : a + b = t\}$, clearing Cayley denominators gives

$$(2 - t)xy + t(x + y) - (2 + t) = 0,$$

an absolutely irreducible bidegree-$(1, 1)$ curve in odd characteristic.

Theorem 3.1 (Direct pointwise Cayley bound)**.** *If*

$$100 < K < \frac{1}{3}p^{3/4},$$

then

$$E_+(A) < 34K^{8/3}. \tag{11}$$

Proof. For $t \neq 0, -2$, the source form of Vyugin–Makarychev's subgroup-curve theorem [13, 14] gives $r_A(t) \leq 32K^{2/3}$ after its degree hypotheses are checked on the displayed bilinear polynomial. The exceptional fibres satisfy $r_A(0) = K - 1$ and $r_A(-2) \leq K - 1$. Since $\sum_t r_A(t) = (K-1)^2$,

$$E_+(A) \leq 2(K-1)^2 + 32K^{2/3} \sum_{t \neq 0,-2} r_A(t) < 34K^{8/3}.$$

□

4 Prime-order rigidity, square resultants and discriminants

Let $\zeta = e^{2\pi i/K}$. Then

$$c(\zeta^a) = -i \cot \frac{\pi a}{K}.$$

Chowla's prime-modulus cotangent independence [8, 9] identifies the characteristic-zero Cayley set with a rational cross-polytope.

Theorem 4.1 (Prime-order additive rigidity)**.** *Every characteristic-zero equal-difference collision among* $\{c(\zeta^a) : a \in \mathbf{F}_K^*\}$ *is identity or central inversion-swap.*

For every admissible cleared collision polynomial $P(X)$, direct algebra gives $P(X) = (X-1)^3 Q(X)$ and a reciprocal relation for Q. Passing to the maximal real subfield yields:

Theorem 4.2 (Square-resultant theorem)**.**

$$\mathrm{Res}(\Phi_K, P) = K^3 M^2, \qquad M \in \mathbf{Z}. \tag{12}$$

For a non-forced characteristic-zero collision, $M \neq 0$.

The Fibonacci slice

$$\mathrm{Res}(\Phi_K(X), X^2 + 3X + 1) = F_K^2$$

shows that individual collision resultants can be exponentially large. At $K = 131$, $p = F_{131}$, a Lucas primality certificate and independent Python/C++ arbitrary-precision computations give exact nonzero Cayley excess 32; this is a finite certificate, not an asymptotic theorem. The primitive-divisor input for the associated rational model is from [12].

Let S_K be the primitive squarefree nonzero-difference polynomial obtained from the resultant of the characteristic-zero Cayley polynomial with its translate. Then for split primes

$$\Delta_K(p) > 0 \iff p \mid \mathrm{Disc}(S_K), \qquad \Delta_K(p) \leq 4v_p(\mathrm{Disc}(S_K)). \tag{13}$$

Ratio-orbit factorization and Dedekind-sum separation [15] give the exact global form

$$|\mathrm{Disc}(S_K)| = 2^{(K-1)^4/4} K^{(K-1)(K^2-7)/4} J_K^4 \tag{14}$$

for an integer $J_K > 0$. Exact computations show that the residual logarithmic mass genuinely remains K^4-scale, so (14) is structural, not a hidden proof of (10).

5 Fixed-product rows and exact split support

For one valid subgroup pair put $u = xy$, $s = x + y$. Then

$$c(x) + c(y) = \frac{2(u-1)}{u-s+1}. \tag{15}$$

At fixed $u \neq 1$ and nonzero parameter there is at most one unordered pair. Let $I_K(p)$ be the maximum number of common parameters of two distinct product rows. The exact cross-row decomposition gives

$$\Delta_K(p) \leq 4I_K(p)(K-1)(K-2). \tag{16}$$

Normalize a row pair to $(1, r)$ and set

$$\tau = \zeta + \zeta^{-1}, \qquad \alpha_r = U_{r-1}(\tau/2), \qquad \beta_r = -2U_{r-2}(\tau/2).$$

The row equation is the affine trace relation

$$Y = \alpha_r X + \beta_r. \tag{17}$$

Let $C_K(T)$ be the minimal polynomial of τ and set

$$H_K(T) = (T-2)C_K(T).$$

The two full row polynomials have one unique characteristic-zero common trace, the invalid row trace. Removing it produces fixed-degree source/target cofactors over $\mathbf{Z}[\tau]$. For every split embedding prime $\mathfrak{q}$,

$$\deg \gcd(P_{K,r}, Q_{K,r} \bmod \mathfrak{q}) = I_{1,r}(\mathfrak{q}). \tag{18}$$

Let the coefficient ideal of the first three fixed-degree subresultants contract to $\Lambda^{(3)}_{K,r}\mathbf{Z}$.

Theorem 5.1 (Exact real-trace order-three support)**.** *For every split rational prime $p \equiv 1 \pmod{K}$,*

$$p \mid \Lambda^{(3)}_{K,r}$$

if and only if some row pair in ratio class $[r]$ has at least three common Cayley parameters.

Proof sketch. The valid cofactors remain fixed-degree and squarefree after split reduction, so (18) is exact. Brown–Traub's fixed-degree subresultant criterion [16] identifies gcd degree at least three with vanishing of the first three subresultants. Their coefficient ideal records simultaneous vanishing in one prime ideal; contraction to $\mathbf{Z}$ gives the rational-prime statement. □

After removing factors $2, K$ and multiplying over ratio classes, write the result as ${\Lambda'_K}^{(3)}$. Then

$$p \nmid {\Lambda'_K}^{(3)} \Longrightarrow I_K(p) \leq 2, \quad \Delta_K(p) \leq 8(K-1)(K-2), \quad E_+(A) \leq 11(K-1)(K-2). \tag{19}$$

The older dehomogenized D'/G' invariants remain valid only as one-way obstructions because specialization can reduce leading degree at projective infinity.

6 Characteristic-zero affine-intercept rigidity

Write

$$t_a = \zeta^a + \zeta^{-a}, \qquad s_a = \zeta^a - \zeta^{-a}, \qquad \gamma_r(a,b) = t_b - \alpha_r t_a.$$

Let $[a]$ denote the trace class $a \sim -a$, and put $E_j = |1-\zeta^j|$, $1 \le j \le m = (K-1)/2$.

Lemma 6.1 (Prime-conductor multiplicative independence)**.** *The positive numbers $E_1, \dots, E_m$ are multiplicatively independent.*

Proof. Bass's prime-power theorem for the group generated by $1-\zeta^j$, modulo torsion, gives a free $\mathbf{Z}$-basis represented by one Galois orbit modulo complex conjugation [10]. Squaring any multiplicative relation among the absolute values turns it into a relation in this torsion-free group, forcing all exponents to vanish. □

Use the elementary identities

$$|s_n| = E_{[2n]}, \qquad |t_x - t_y| = E_{[x-y]}E_{[x+y]}. \tag{20}$$

Theorem 6.2 (Trace-intercept fibre rigidity)**.** *For every odd prime $K \ge 7$ and every normalized ratio $r \ne 1$, every fibre of $([a],[b]) \mapsto \gamma_r(a,b)$ has cardinality at most two.*

If $r = -1$, a double fibre is only a swap. If $r \ne \pm 1$, every double fibre is, up to signs and exchange,

$$([a],[a+1-r]), \qquad ([a+2],[a+1+r]). \tag{21}$$

Proof. Suppose two distinct pairs have the same intercept. For $r \ne -1$,

$$t_b - t_d = \alpha_r(t_a - t_c).$$

Taking absolute values and applying (20) yields

$$E_{[b-d]}E_{[b+d]}E_{[2]} = E_{[a-c]}E_{[a+c]}E_{[2r]}.$$

Lemma 6.1 makes the multisets of folded indices equal. Since $[2r] \ne [2]$, one source difference must be $[2]$; after changing signs and ordering take $c = a+2$. The remaining two indices force the target classes in (21). The equality itself follows from

$$t_{x+r} - t_{x-r} = s_r s_x, \qquad t_{a+2} - t_a = s_1 s_{a+1}.$$

A third point would form a triangle in the step-two quotient of the odd K-cycle by $a \sim -a$; that quotient is a path. For $r = -1$ the intercept is $t_a + t_b$, and prime cyclotomic linear independence leaves only identity or swap. □

The distinguished characteristic-zero row solution $([1],[r])$ is therefore isolated. This mechanism is related to cyclotomic-unit cross-ratio methods in projective equivalence of roots of unity [11], but the affine theorem above is proved directly in the prime-conductor setting.

7 Cyclotomic-unit matching and a scalar shift resultant

Put $u_n = s_n/s_1$. For valid trace classes define

$$W_r(b) = u_{(b+r)/2}u_{(b-r)/2}, \qquad V_r(a) = u_r u_{(a+1)/2}u_{(a-1)/2}. \tag{22}$$

A direct difference factorization gives

$$\gamma_r(a,b) - \beta_r = s_1^2(W_r(b) - V_r(a)). \tag{23}$$

Since s_1 is a unit away from K, every split row intersection is exactly a congruence $W_r(b) \equiv V_r(a) \pmod{\mathfrak{q}}$ between explicit products of cyclotomic units.

Let

$$P_1(T) = \frac{H_K(T)}{T - t_1}, \qquad R_r(U) = \frac{H_K(U)}{U - t_r}$$

and define the scalar shift resultant

$$\mathcal{R}_{K,r}(Z) = \mathrm{Res}_T(P_1(T), R_r(\alpha_r T + \beta_r + Z)) = c_0 + c_1 Z + c_2 Z^2 + \cdots . \tag{24}$$

Theorem 7.1 (Scalar support identity)**.** *For every split embedding prime* $\mathfrak{q}$*,*

$$\mathrm{ord}_{Z=0}(\mathcal{R}_{K,r}(Z) \bmod \mathfrak{q}) = I_{1,r}(\mathfrak{q}). \tag{25}$$

Consequently

$$I_{1,r}(\mathfrak{q}) \geq 3 \iff c_0 \equiv c_1 \equiv c_2 \equiv 0 \pmod{\mathfrak{q}}.$$

Proof. Over the split residue field the two valid trace polynomials factor linearly, so the product formula gives

$$\mathcal{R}_{K,r}(Z) = \prod_{[a] \neq [1]} \prod_{[b] \neq [r]} (Z + \alpha_r t_a + \beta_r - t_b).$$

Each factor is linear in Z and vanishes at zero exactly for one row intersection. □

The ideal (c_0, c_1, c_2) contracts to an integer $\Omega^{(3)}_{K,r}$ with the same split rational-prime order-three support as $\Lambda^{(3)}_{K,r}$. Equality of the full integers is not asserted. Moreover, the nonzero characteristic-zero constant coefficient gives the per-ratio deterministic budget

$$\sum_{\substack{p \equiv 1 \ (K) \\ I_r(p) \geq 3}} \log p = O(K^3). \tag{26}$$

Summing over $O(K)$ ratio classes recovers an $O(K^4)$ scale, so (26) is not a polynomial largest-prime theorem.

8 Inverse reflection and the pair-sum square

The ratio $r = -1$ has additional symmetry. Here

$$\alpha = -1, \qquad \beta = 2\tau, \qquad Q(T) = -P(2\tau - T).$$

Set

$$F(X) = P(\tau + X) = E(X^2) + XO(X^2). \tag{27}$$

The reflection reduction gives

$$I_{-1}(\mathfrak{q}) = 2 \deg \gcd(E, O \bmod \mathfrak{q}). \tag{28}$$

Thus inverse-row intersections are automatically even. A fixed-degree half-size subresultant construction defines the exact split support integer Ξ_K.

The scalar shift resultant admits a stronger universal factorization.

Theorem 8.1 (Inverse pair-sum factorization)**.** *Let* $x_1, \ldots, x_m$ *be the roots of the monic polynomial* F*. Define*

$$S(Z) = \prod_{i<j} (Z - x_i - x_j).$$

Then $S \in \mathbf{Z}[\tau][Z]$ and

$$\mathrm{Res}_X(F(X), F(Z-X)) = 2^m F(Z/2) S(Z)^2. \tag{29}$$

For every split embedding prime,

$$I_{-1}(\mathfrak{q}) = 2\,\mathrm{ord}_{Z=0}(S(Z) \bmod \mathfrak{q}). \tag{30}$$

Hence

$$I_{-1}(\mathfrak{q}) \geq 3 \iff S(0) \equiv S'(0) \equiv 0 \pmod{\mathfrak{q}}. \tag{31}$$

Proof. The root-product formula gives

$$\mathrm{Res}_X(F(X), F(Z-X)) = \prod_{i,j}(Z - x_i - x_j).$$

The diagonal ordered pairs give $2^m F(Z/2)$, and every unordered pair $i < j$ occurs twice. The coefficients of S are symmetric integral polynomials in the roots. At a split prime, $F(0) = P(\tau) = H'_K(\tau)$ is nonzero modulo every embedding prime, so the diagonal factor is a unit at $Z = 0$. Combining with the general shift-resultant identity gives (30); (31) follows because the left side is even. □

Let $S(Z) = s_0 + s_1 Z + \cdots$ and let the ideal (s_0, s_1) contract to $\Theta_K \mathbf{Z}$. Then Θ_K, Ξ_K, and the inverse factor of $\Lambda^{(3)}_{K,-1}$ have the same split rational-prime support; their valuations need not agree.

9 The pair-sum polynomial as a translated half-degree resultant

The square root S itself can be obtained without constructing the degree-m^2 shift resultant. Translate

$$F_Z(X) = F(X + Z/2) = E_Z(X^2) + X O_Z(X^2). \tag{32}$$

Theorem 9.1 (Translated parity-resultant identity)**.** *There exists a convention-dependent sign $\varepsilon_m \in \{\pm 1\}$ such that*

$$S(Z) = \varepsilon_m \,\mathrm{Res}_Y(E_Z(Y), O_Z(Y)). \tag{33}$$

Consequently

$$S(0) = \varepsilon_m \,\mathrm{Res}_Y(E, O), \tag{34}$$

$$S'(0) = \varepsilon_m \,\frac{d}{dZ}\,\mathrm{Res}_Y(E_Z, O_Z)\Big|_{Z=0}. \tag{35}$$

The first-order deformation is

$$\partial_Z E_Z|_0 = \frac{O + 2YO'}{2}, \qquad \partial_Z O_Z|_0 = E'. \tag{36}$$

Proof. The roots of F_Z are $y_i = x_i - Z/2$. Apply the constant pair-sum identity from Theorem 8.1 to the monic polynomial F_Z. Its unordered pair-sum product at zero is

$$\prod_{i<j}(-y_i - y_j) = \prod_{i<j}(Z - x_i - x_j) = S(Z).$$

The even/odd decomposition (32) identifies this product, up to the fixed resultant sign, with $\mathrm{Res}(E_Z, O_Z)$. Differentiating $F(X + Z/2)$ at $Z = 0$ gives (36). □

Thus inverse order-three support is the simultaneous divisibility problem for a half-degree resultant and its translation derivative. In exact computation, both scalars can be obtained in one Sylvester determinant over dual numbers $\varepsilon^2 = 0$, reducing the determinant dimension to $O(K)$ rather than materializing a degree-m^2 polynomial.

Exact Smith contractions give, after removing factors $2K$,

$$\Theta'_7 = 1, \quad \Theta'_{11} = 1, \quad \Theta'_{13} = 5, \quad \Theta'_{17} = 1, \quad \Theta'_{19} = 37, \quad \Theta'_{23} = 47^2 \cdot 137.$$

For comparison $\Xi'_{23} = 47 \cdot 137$: split prime support agrees, valuations differ. The reference SymPy implementation reduces the $K = 17$ computation from roughly 51 seconds for the full shift resultant to roughly 0.6 seconds for the translated half-degree route; at $K = 23$ the latter remains on the order of seconds, after which Smith normal form becomes the principal finite algebra bottleneck. These timings are reproducibility data, not asymptotic complexity theorems.

The norm of the inverse constant shift coefficient also has the exact shape

$$|\mathrm{N}(R_-(0))| = 2^{m^2} K^m \, |\mathrm{N}(\mathrm{Res}(E, O))|^2 , \tag{37}$$

using $|\mathrm{N}(F(0))| = K^m$. Hence every prime away from $2K$ occurs with even valuation in this particular norm. This explains part of the observed square pattern but does not imply equality of the support integers.

10 Global split-prime budget and minimal-extension Frobenius

Simultaneous multiplication of collision exponent indices by $a \in \mathbf{F}_K^\times$ acts freely on non-forced collision quadruples. The number of normalized scaling orbits is

$$N_K = K^3 - 5K^2 + 6K + 1.$$

Combining the square-resultant theorem with $|\mathrm{Res}(\Phi_K, P)| \le 12^{K-1}$ gives:

Theorem 10.1 (Weighted split-prime budget)**.**

$$\sum_{p \equiv 1 \,(K)} \Delta_K(p) \log p \le N_K\big((K-1)\log 12 - 3\log K\big). \tag{38}$$

For prime $K \asymp X^\theta$ and split prime $p \asymp X$, Bombieri–Vinogradov and PNT [17, 18] imply

$$\text{average } \frac{\Delta_K(p)}{K^2} \ll_\theta X^{3\theta-1}$$

for fixed $\theta < 1/3$. Integrality of the excess strengthens this to a density-one exact baseline for fixed $\theta < 1/5$:

$$\Delta_K(p) = 0, \qquad E_+(A) = 3(K-1)(K-2)$$

for asymptotically almost every split pair.

Now let $p \neq 2, K$ be arbitrary and put

$$f = \mathrm{ord}_K(p), \qquad q = p^f.$$

In $\mathbf{F}_q$ the order-K subgroup exists uniquely.

Theorem 10.2 (Frobenius amplification)**.** *Every non-forced collision orbit under the p-power Frobenius has size exactly f. Hence*

$$f \mid \Delta_K^{\mathrm{ext}}(p).$$

The weighted resultant budget and the discriminant classification extend to all such rational primes in the minimal extension.

Proof. Frobenius multiplies every exponent by p modulo K. If a positive power $d < f$ fixed a collision tuple, any nonzero coordinate would imply $p^d \equiv 1 \pmod{K}$, contradicting the definition of f. The resultant argument is applied at one prime ideal over p; $p \neq K$ is unramified, so its valuation on rational integers is the ordinary p-valuation. □

PNT, Brun–Titchmarsh and the scarcity of small multiplicative orders then give a density-one exact baseline for unrestricted prime pairs $K \asymp X^\theta$, $p \asymp X$ for every fixed $\theta < 1/3$. This mechanism is absent on the split slice because there $f = 1$.

11 Full Cayley–Jacobi trace and corrected off/off geometry

Let $S = H^\perp$ be the multiplicative-character annihilator and define the normalized twisted-Jacobi matrix

$$M_S(q) = \frac{e_p(-q/2)}{p-1} J_S(q).$$

Character orthogonality gives the exact trace identity

$$\widehat{1_A}(r) = K \operatorname{Tr} M_S(2r), \tag{39}$$

and therefore

$$\sum_q |\operatorname{Tr} M_S(q)|^2 = \frac{p(K-1)}{K^2}, \tag{40}$$

$$\sum_q |\operatorname{Tr} M_S(q)|^4 = \frac{pE_+(A)}{K^4}. \tag{41}$$

The normalized fourth/second moment ratio tends to 3 in the maintained split double average below $\theta < 1/3$. This is a fourth-moment theorem, not a CLT.

The pure off/off family contains a coherent reducible fibre. For generic $\lambda = (h_1, h_2)$ it occurs at

$$t_\lambda = \frac{2}{c(h_1) - c(h_2)}$$

and has exactly $2p-8$ valid affine points. After extracting the spike, Igusa geometric monodromy [20] and the Fouvry–Kowalski–Michel trace-correlation argument [19] give square-root quasi-orthogonality between non-isomorphic geometric parameter classes.

Theorem 11.1 (Corrected off/off variance)**.**

$$\operatorname{Var}_{t \neq 0} O(t) \ll pE_+(A) + K^2 p + K^4 p^{1/2} + K^4, \tag{42}$$

with an absolute implied constant.

Thus the off/off layer has collapsed onto the same pointwise Cayley-energy bottleneck as the mixed layer.

12 Exact trace-to-local-statistics bridge

Let U be an $N \times N$ unitary matrix with eigenvalues $e^{2\pi i\theta_1}, \dots, e^{2\pi i\theta_N}$. For a zero-sum integer tuple $m_1 + \cdots + m_k = 0$, define

$$D_U(m_1, \dots, m_k) = \sum_{j_1, \dots, j_k \text{ distinct}} e^{2\pi i \sum_{a=1}^k m_a \theta_{j_a}}.$$

For a set partition π of $\{1, \dots, k\}$, put

$$\mu(\pi) = \prod_{B \in \pi} (-1)^{|B|-1} (|B| - 1)!.$$

Theorem 12.1 (Factorial trace identity)**.**

$$D_U(m_1,\dots,m_k)=\sum_{\pi}\mu(\pi)\prod_{B\in\pi}\operatorname{Tr}U^{\sum_{a\in B}m_a}. \tag{43}$$

This is Möbius inversion on the partition lattice. Let $\phi\in\mathcal{S}(\mathbf{R}^{k-1})$ and periodize it at the mean-spacing scale. Poisson summation gives the exact microscopic statistic

$$C_{k,N}(U;\phi)=N^{-k}\sum_{m\in\mathbf{Z}^{k-1}}\widehat{\phi}(m/N)D_U(m_1,\dots,m_k), \tag{44}$$

where $m_k=-(m_1+\cdots+m_{k-1})$. In particular,

$$C_{2,N}(U;\phi)=N^{-2}\sum_{m\in\mathbf{Z}}\widehat{\phi}(m/N)(|\operatorname{Tr}U^m|^2-N). \tag{45}$$

If $\widehat{\phi}$ has compact support, the relevant powers satisfy $|m|=\Theta(N)$: microscopic local statistics probe the Heisenberg scale.

A weighted comparison of the quantities $D_U(m)$ with the matching random matrix ensemble on this range is therefore sufficient for the corresponding band-limited microscopic correlation theorem. Arbitrary Schwartz tests additionally require a uniform high-frequency tail bound. This deterministic Fourier bridge is exact; the required arithmetic/spectral joint-trace estimates for the correct desymmetrized cat-map block remain open.

13 Reproducibility and adversarial controls

The maintained repository uses distinct failure modes:

- Python/SymPy regenerates exact resultants, discriminants, Smith contractions, finite-extension Frobenius checks, trace identities and the translated inverse-row support;
- C++20 independently checks finite-field hot paths, arbitrary-precision witnesses and characteristic-zero trace-intercept fibres;
- Lean 4 kernel-checks maintained finite certificates and falsifiers;
- CI rebuilds all maintained code and recompiles this article; a separate workflow synchronizes the committed PDF from the TeX source.

The C++ trace-intercept verifier checks every normalized ratio through the maintained quick range $K\le 199$; a recorded extended run reaches prime $K=809$, always with maximum fibre size two and the predicted double-fibre form. The inverse translated-resultant verifier has exact Smith anchors through $K=23$. Representative prime-to-$2K$ contractions are

$$\Theta'_{13}=5,\qquad \Theta'_{19}=37,\qquad \Theta'_{23}=47^2\cdot 137.$$

The independent reflection invariant gives $\Xi'_{23}=47\cdot 137$, confirming equal split prime support but different valuation. No integer equality is inferred.

Adversarial finite controls include $(K,p)=(29,728017)$ and $(31,1042903)$ with $p>K^4$ but positive excess, disproving a bare "$p>K^4$ implies baseline" rule; $(K,p)=(41,2908049)$ has a row pair with two common parameters, disproving $I_K(p)\le 1$; and $(K,p)=(31,1303)$ has inverse-row intersection size four.

14 Fixed-map Pell blocks: a global Gaussian law and its rate

Consider the fixed hyperbolic map and associated constants

$$T = \begin{pmatrix} 3 & 2 \\ 4 & 3 \end{pmatrix}, \qquad \alpha = 1 + \sqrt{2}, \qquad \lambda = \alpha^2.$$

For each odd prime K, set $N + F\sqrt{2} = \alpha^K$, so that $N^2 - 2F^2 = -1$. The modulus N may be composite. Put $s = F^{-1} \pmod N$, $h = 3 + 2s \pmod N$ and $q = -F \pmod N$. Then $s^2 = 2$, and h has exact order K modulo each prime divisor of N. In particular, $h^d - 1$ is a unit for $0 < d < K$, and $h^d + 1$ is a unit for all d.

Use Weyl operators

$$(W(a,b)f)(x) = e_N(a(x - b/2))f(x - b), \qquad (S_h f)(x) = f(h^{-1}x),$$
$$Q_j = \frac{1}{K}\sum_{t=0}^{K-1} e^{-2\pi i j t/K} S_h^t.$$

The explicit symplectic diagonalization in the maintained Pell note takes the original cosine observable to the odd-sector blocks

$$A_j = Q_j\{W(q,1) + W(q,1)^*\}Q_j|_{\mathrm{odd}}\,, \qquad n = \frac{N-1}{2K}, \qquad X_j = \sqrt{K/2}\,A_j.$$

Each sector has dimension n, including at composite moduli. The symmetrized observable preserves parity; an individual Weyl translation need not do so. All claims here concern these observable blocks.

Theorem 14.1 (Global Gaussian limit and rate)**.** *For every fixed integer $r \geq 1$, as K tends to infinity through odd primes,*

$$\sup_{0 \leq j < K} \left| n^{-1} \operatorname{Tr} X_j^{2r} - (2r-1)!! \right| = O_r(K^{-1}) + O_r(K^{r+1}/N).$$

For each fixed odd m and all sufficiently large K,

$$\sup_j |n^{-1} \operatorname{Tr} X_j^m| \leq 2^{m/2+1} K^{m/2+1}/(N-1).$$

Thus the global empirical measures converge weakly to the variance-one Gaussian, uniformly in the sequential choice of sector.

Proof. The fixed-point Weyl trace identity separates central reciprocal-zero words from a remainder at most $2^{m/2+1}K^{m/2+1}/(N-1)$. The unit conditions justify this estimate at composite N.

A modular reciprocal relation with m signed terms can be rotated over its largest cyclic gap. Its polynomial has degree at most $K - K/m$ and coefficient ℓ^1 norm at most m. If its value at λ were nonzero, its algebraic norm would be a nonzero multiple of N^2 but have absolute value at most $m^2\lambda^{K-K/m}$. Since $N^2 \asymp \lambda^K$, this is impossible for fixed m and large K. Hence the relation lifts to characteristic zero. The minimal polynomial $x^2 - 6x + 1$ takes value -4 at 1, excluding odd signed relations. The same norm argument shows that a gap exceeding $2\log_\lambda m$ splits a relation. Bounded-width components then give $O_r(K^{r-2})$ pinned nonpairable configurations, compared with K^{r-1} paired configurations. Detailed component counting is in the Pell notes.

It remains to control crossing phases without a logarithmic loss. Let P_t be the Pell sequence, and let δ be the signed representative of d modulo K. Exactly,

$$z_d = e_N(q(h^d - h^{-d})) = e_N(-2P_{2\delta}).$$

Writing $\delta = (K-1)/2 - \ell$ for positive δ gives

$$\frac{P_{2\delta}}{N} = \frac{\lambda^{-\ell}}{\sqrt{2}\alpha} \frac{1-\lambda^{-2\delta}}{1-\lambda^{-K}}.$$

Since $\alpha(1-\lambda^{-1}) = 2$, summing $|e^{it}-1| \le |t|$ yields

$$\sum_{d \bmod K} |z_d - 1| \le \frac{2\sqrt{2}\pi}{1-\lambda^{-K}} < 9.$$

Each crossing label difference is uniform under pairing-label averaging, even when one pair label is pinned. The product inequality for unit phases therefore gives a mean error at most $9/K$ per crossing, without requiring independence of different crossings. Repeated-label overcount and nonpairable words contribute $O_r(K^{-1})$. The stated trace bounds follow. Gaussian moment determinacy and higher even-moment bounds give weak convergence. □

Theorem 14.2 (First fourth-moment correction)**.** *For odd prime $K \ge 11$, uniformly in j,*

$$n^{-1} \operatorname{Tr} X_j^4 = 3 - \frac{\gamma}{K} + O(K^3/N),$$

where

$$\gamma = \frac{3}{2} + 2\sum_{\ell \ge 0} \left[1 - \cos\left(\frac{4\pi}{\sqrt{2}\alpha}\lambda^{-\ell}\right)\right] = 5.61422069215\ldots.$$

The following finite formula has an explicit remainder:

$$n^{-1} \operatorname{Tr} X_j^4 = \frac{N}{N-1}\left(2 + S_K - \frac{3}{2K}\right) + R_{4,j}, \qquad |R_{4,j}| \le \frac{8K^3}{N-1},$$

$$S_K = \frac{1}{K}\left(1 + 2\sum_{\delta=1}^{(K-1)/2} \cos(4\pi P_{2\delta}/N)\right).$$

Proof. Four-term reciprocal modular relations lift for $K \ge 11$, since $\lambda^{K/4}(1-\lambda^{-K})^2 > 64$. A lifted four-term relation pairs because $\lambda > 3$. Two pairings are noncrossing and one is crossing. The six balanced all-equal sign words are counted twice; subtracting that overcount gives $3/(2K)$. The noncentral fixed-point trace estimate gives $R_{4,j}$. Reindexing the exact Pell ratio yields the cosine series. The summed finite-argument error is $O(N^{-1})$ before division by K, and the omitted cosine tail is $O(\lambda^{-K})$. With $a = 4\pi/(\sqrt{2}\alpha)$, the nonnegative tail after J terms is at most $a^2\lambda^{-2J}/(1-\lambda^{-2})$. □

Verification and scope. The independent orbit-matrix verifier includes nontrivial sectors and $N = 8119 = 23 \cdot 353$. Its matrix-power and eigenvalue moments are numerical checks, not replacements for the proofs. The displayed decimal for γ is now independently enclosed by pure-integer interval arithmetic: Machin arctangent bounds, integer square root, outward arithmetic, cosine Taylor remainders, and a geometric tail. The separate matrix diagnostics still use floating-point arithmetic. Existing Lean certificates concern explicit finite word traces; they do not formalize these general limit theorems. Priority remains unverified. Neither this Gaussian global density nor an exact all-order trace formula establishes local spacings or Heisenberg-scale universality for the propagator.

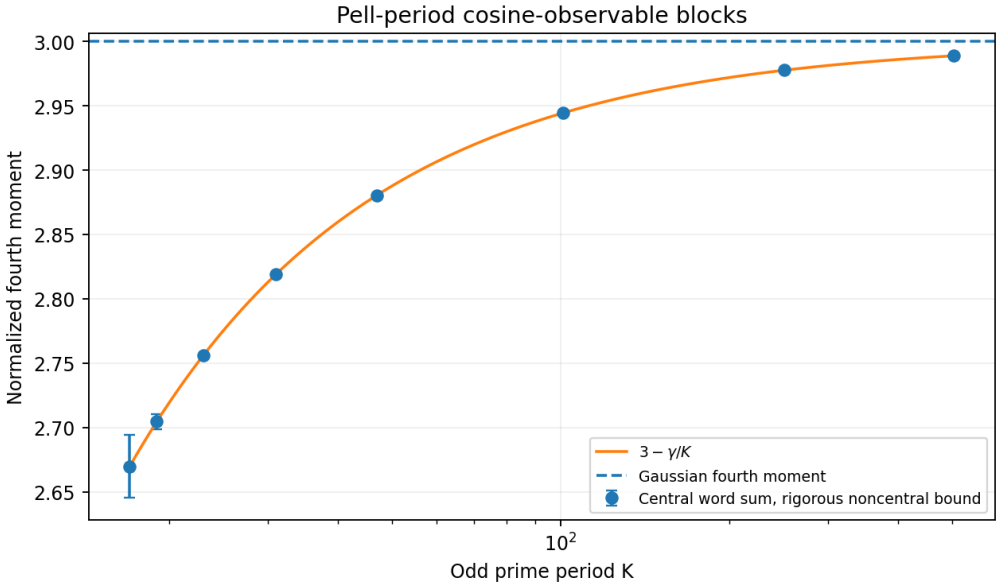


Figure 1: Central fourth-moment word sums with the explicit noncentral bound, compared with $3 - \gamma/K$. This is not a level-spacing plot.

15 An arithmetic normal form for the actual kicked propagator

The Pell observable limit extends to an explicitly controlled spectral law of the perturbed propagator at its recurrence time. This extension concerns folded eigenphases, not microscopic BGS statistics. A self-contained proof, including a direct odd-space Gaussian moment argument, is supplied in the companion *A resonance-scale spectral law for a deterministically kicked Pell cat map* (`pell_floquet_normal_form.pdf`).

Let $K \geq 11$ be an odd prime, $N + F\sqrt{2} = (1+\sqrt{2})^K$, $h = 3 + 2F^{-1} \pmod N$, $q = -F \pmod N$, and $d = (N-1)/2$. In the odd subspace set

$$U_0 = S_h^{-1}, \quad B = W(q,1) + W(q,1)^*, \quad U_\varepsilon = e^{i\varepsilon B} U_0, \quad \overline{B} = \frac{1}{K} \sum_{j=0}^{K-1} U_0^j B U_0^{-j}.$$

This entire kicked family is unitarily equivalent to a fixed cosine observable kicking the fixed map $T = \left(\begin{smallmatrix} 3 & 2 \\ 4 & 3 \end{smallmatrix}\right)$. The equivalence does not depend on primality of N.

Theorem 15.1 (Finite normal form and critical recurrence law)**.** *Put $B_j = U_0^j B U_0^{-j}$, $c_j = (K-1-2j)/(2K)$, $G = \sum_j c_j B_j$ and $V_\varepsilon = e^{-i\varepsilon G}$. For every real ε,*

$$\left\| V_\varepsilon U_\varepsilon V_\varepsilon^* - e^{i\varepsilon \overline{B}} U_0 \right\|_{2,-} \leq \frac{12\varepsilon^2}{\sqrt{K}}, \qquad \|A\|_{2,-}^2 = d^{-1} \operatorname{Tr}_-(A^* A).$$

For each fixed real τ, $\varepsilon = \tau/\sqrt{2K}$ and each integer m,

$$\frac{1}{d} \operatorname{Tr}_- U_\varepsilon^{mK} \longrightarrow e^{-m^2\tau^2/2}.$$

Equivalently, the empirical eigenvalue measures of U_ε^K converge to the wrapped normal law of $e^{i\tau Z}$, $Z \sim N(0,1)$.

Mechanism of the estimate. The centered coefficients solve $G - U_0 G U_0^* = B - \overline{B}$ exactly. For $a_j = |e_N(q(h^j - h^{-j})) - 1|$ and $\lambda = 3 + 2\sqrt{2}$, Pell summation gives

$$\sum_j a_j < 9, \qquad \sum_j |c_j| a_j \le \frac{2\pi}{K(1 - \lambda^{-K})}.$$

Four-letter quadratic lifting makes all distinct-time signed pair sums of Weyl indices different. Orthogonality in the *full* Weyl space gives

$$\|[G, \overline{B}]\|_{2,N}^2 = \frac{4}{K^4} \sum_{j=1}^{K-1} (K - j) j^2 a_j^2 \le \frac{9}{K}.$$

Also $\|[G, B]\|_{\mathrm{op}} < 26/K$ and $\|[B, \overline{B}]\|_{\mathrm{op}} < 36/K$. For Hermitian A, D, Duhamel's integral identity yields

$$\|e^{i\varepsilon A} e^{i\varepsilon D} - e^{i\varepsilon(A+D)}\| \le \tfrac{1}{2}\varepsilon^2 \|[A, D]\|$$

in either relevant unitarily invariant norm. Applied twice, this proves the stated constant without requiring $\varepsilon\|G\|$ to be small. Telescoping mK powers and using $U_0^K = I$ leaves comparison error $6|m|\tau^2/\sqrt{K}$ against $d^{-1} \operatorname{Tr}_- \exp(im\tau X_K)$, where $X_K = \sqrt{K/2}\,\overline{B}$. The Gaussian moment law for X_K proves the limit. The comparison rate is not asserted as the total rate of Gaussian characteristic-function convergence. □

Corollary 15.2 (Subcritical folded phases)**.** *If $\varepsilon_K \neq 0$ and $\tau_K = \varepsilon_K \sqrt{2K} \to 0$, then*

$$\frac{\operatorname{Arg}(e^{iK\theta_\ell})}{\tau_K}, \qquad e^{i\theta_\ell} \in \operatorname{Spec}(U_{\varepsilon_K}),$$

have empirical measures converging in W_2 to $N(0, 1)$. No labelling into isolated old clusters is required. The proof bounds the additional matching error by $24|\tau_K|/\sqrt{K} + (6|\tau_K|/\pi)\sqrt{d^{-1} \operatorname{Tr}_- X_K^4}$.

The quantum kick at criticality tends to zero; its corresponding classical strength is of order $1/(N\sqrt{K})$. The recurrence time $K \asymp \log N$ is much shorter than the Heisenberg scale d. These statements do not establish local spacing universality or the fixed-classical-perturbation BGS conjecture. Priority for the specific arithmetic construction remains unverified. Independent C++ and Python representations, finite Lean kernels, and exact-versus-sampled numerical provenance are maintained separately.

16 Remaining frontier

The arithmetic fourth-moment chain is now

common split divisors of small cyclotomic scalar systems

↓

pointwise centered Cayley energy

↓

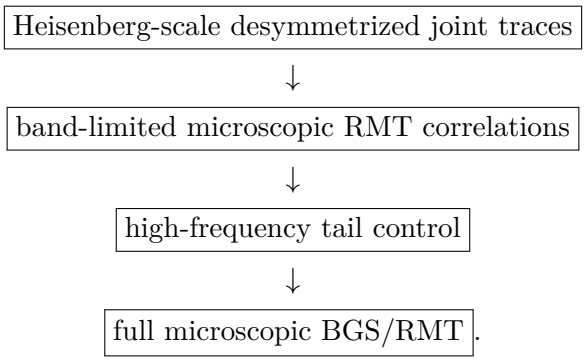
complete fourth-moment control.

For general ratios the order-three event is encoded by three scalar shift coefficients; for the inverse ratio it reduces further to a half-degree resultant and its translation derivative. The existing per-ratio $O(K^3)$ logarithmic mass does not bound the largest supported split prime polynomially. A sufficient remaining target is, for example,

$$P^+_{\mathrm{split}}(\Lambda'_K{}^{(3)}) \le K^{4+o(1)},$$

or an equivalent theorem for the scalar support systems. The high-value next step is a uniform common-divisor/Bezout theorem, not another individual height estimate.

The microscopic spectral chain is independently



The Fourier arrow is exact by (43)–(44); the joint-trace estimate itself is open. Fixed powers, finitely many moments, or even a completed fourth-moment theorem cannot be promoted to microscopic BGS.

17 Conclusion

The fourth-moment problem has been reduced to a progressively smaller arithmetic obstruction. Fixed-product rows give an exact order-three split support. Prime-conductor cyclotomic-unit rigidity closes the corresponding characteristic-zero fibre problem. A scalar shift resultant compresses the general finite event to three coefficients, while the inverse row admits an additional perfect-square factorization that reduces the event to two half-degree scalar quantities: a resultant and its translation derivative. These reductions are exact and specialization-safe, but they do not yet supply a polynomial largest-split-prime theorem.

Frobenius amplification yields a complementary density-one exact theorem over minimal finite extensions while correctly leaving the split slice unchanged. On the spectral side, the exact partition-Möbius/Poisson identity identifies Heisenberg-scale joint traces as the missing input for microscopic correlations. Full BGS/GOE/COE universality remains open.

References

[1] P. Kurlberg and Z. Rudnick, *On the distribution of matrix elements for the quantum cat map*, Ann. of Math. **161** (2005), 489–507, doi:10.4007/annals.2005.161.489.

[2] P. Kurlberg, L. Rosenzweig and Z. Rudnick, *Matrix elements for the quantum cat map: fluctuations in short windows*, arXiv:math/0701685 (2007).

[3] S. Dyatlov and M. Jézéquel, *Semiclassical Measures for Higher-Dimensional Quantum Cat Maps*, Ann. Henri Poincaré **25** (2024), 1545–1605, doi:10.1007/s00023-023-01309-x.

[4] P. Kurlberg, A. Ostafe, Z. Rudnick and I. E. Shparlinski, *On Quantum Ergodicity for Higher Dimensional Cat Maps*, Commun. Math. Phys. **406** (2025), 174, doi:10.1007/s00220-025-05350-1.

[5] S. Bhakta and I. E. Shparlinski, *On quantum ergodicity for higher-dimensional cat maps modulo prime powers*, Bull. London Math. Soc. **58** (2026), doi:10.1112/blms.70300.

[6] Z. Pluhař and H. A. Weidenmüller, *Universal Quantum Graphs*, Phys. Rev. Lett. **112** (2014), 144102, doi:10.1103/PhysRevLett.112.144102.

[7] Z. Pluhař and H. A. Weidenmüller, *Quantum graphs and random-matrix theory*, J. Phys. A **48** (2015), 275102, doi:10.1088/1751-8113/48/27/275102.

[8] S. Chowla, *The nonexistence of nontrivial linear relations between the roots of a certain irreducible equation*, J. Number Theory **2** (1970), 120–123, doi:10.1016/0022-314X(70)90012-0.

[9] K. Wang, *On a theorem of S. Chowla*, J. Number Theory **15** (1982), 1–4, doi:10.1016/0022-314X(82)90078-6.

[10] H. Bass, *Generators and Relations for Cyclotomic Units*, Nagoya Math. J. **27** (1966), 401–407, doi:10.1017/S0027763000026258.

[11] H. Fu, *Projective Equivalence for the Roots of Unity*, J. Algebraic Combin. **56** (2022), 823–871, arXiv:1905.01510.

[12] M. Yabuta, *A Simple Proof of Carmichael's Theorem on Primitive Divisors*, Fibonacci Quart. **39** (2001), 439–443, doi:10.1080/00150517.2001.12428701.

[13] I. V. Vyugin and S. Makarychev, *Solutions of polynomial equation over* $\mathbf{F}_p$ *and new bounds of additive energy*, arXiv:1504.01354v2 (2015).

[14] S. Makarychev and I. Vyugin, *Solutions of Polynomial Equations in Subgroups of* $\mathbf{F}_p^*$, Arnold Math. J. **5** (2019), 105–121, doi:10.1007/s40598-019-00112-z.

[15] S. Jabuka, S. Robins and X. Wang, *When are two Dedekind sums equal?*, Int. J. Number Theory **7** (2011), 2197–2203, arXiv:1103.0370.

[16] W. S. Brown and J. F. Traub, *On Euclid's Algorithm and the Theory of Subresultants*, J. ACM **18** (1971), 505–514, doi:10.1145/321662.321665.

[17] E. Bombieri, *On the large sieve*, Mathematika **12** (1965), 201–225, doi:10.1112/S0025579300005313.

[18] H. Iwaniec and E. Kowalski, *Analytic Number Theory*, AMS Colloquium Publications **53**, 2004, doi:10.1090/coll/053.

[19] É. Fouvry, E. Kowalski and P. Michel, *Counting sheaves using spherical codes*, Math. Res. Lett. **20** (2013), 305–323, doi:10.4310/MRL.2013.V20.N2.A8, arXiv:1210.0851.

[20] A. Bandini, I. Longhi and S. Vigni, *Torsion points on elliptic curves over function fields and a theorem of Igusa*, Expo. Math. **27** (2009), 175–209, doi:10.1016/j.exmath.2008.06.001, arXiv:0804.1425.